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TRAINING AND EVALUATION OUTLINE

T&EO-MSS

USAREUR-AF

TRAINING AND EVALUATION OUTLINES (T&EO)

Maven Smart System (MSS) Training Program

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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11 MARCH 2026

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TRAINING AND EVALUATION OUTLINES (T&EO)

MAVEN SMART SYSTEM (MSS) TRAINING PROGRAM

USAREUR-AF Operational Data Team — C2DAO

Document	Training and Evaluation Outlines
Proponent	USAREUR-AF C2DAO
Reference POI	POI-MSS-001
Reference MTP	MTP-MSS-001
Effective Date	March 2026
Classification	UNCLASSIFIED

HOW TO USE T&EOS

Each T&EO defines the conditions and standards for a specific evaluated task. The evaluator uses these outlines to: 1. Brief the trainee on the task conditions before the evaluation 2. Score each performance measure as GO or NO-GO 3. Determine overall task result

Scoring: - Any step marked **[CRITICAL]** = automatic NO-GO if failed, regardless of other performance - All other steps: the evaluator applies GO/NO-GO holistically, considering whether the trainee can perform the task to standard independently

PART I — TM-10: MAVEN USER T&EOS

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T&EO TM10-01: LOG IN AND NAVIGATE TO DESIGNATED APPLICATION

Task: Authenticate to MSS Training Environment and navigate to a designated Workshop application.

Conditions: Given a government workstation with CAC reader and network access to the MSS Training Environment; trainee has an active account in the Training Environment. The evaluator specifies the name of the target Workshop application.

Standards: The trainee will authenticate using CAC/PIV PIN and navigate to the designated Workshop application — without instructor assistance — within 5 minutes of the task being assigned.

#	Performance Measure	GO	NO-GO
1	Opens browser and navigates to the MSS Training Environment URL (not production MSS)	Navigates directly to correct URL	Opens production MSS; requires correction
2	Selects correct certificate (PIV Authentication)	Selects correct cert	Selects wrong cert; authentication fails
3	Enters PIV PIN when prompted	PIN entered correctly	Fails PIN entry; requires reset
4	Confirms Training Environment landing page is displayed	Training Environment confirmed	Navigates to production environment
5	Navigates to designated Workshop application using Projects or Compass	Application open within 5 minutes	Requires instructor guidance; exceeds 5 minutes
[CRITICAL] 6	Does not attempt to access the production MSS environment during the task	Training Environment confirmed	Navigates to production

T&EO TM10-02: FILTER TABLE TO IDENTIFY MISSING SUBMISSIONS

Task: Apply a date filter to a SITREP submission table and identify units with missing submissions.

Conditions: Given an open Workshop application containing a SITREP submission table with at least 20 records spanning multiple weeks; the evaluator specifies the filter condition ("filter to last 7 days") and the unit list (15 units expected).

Standards: The trainee will apply the correct filter and correctly identify all units with missing submissions — without assistance — within 5 minutes.

#	Performance Measure	GO	NO-GO
1	Locates the filter control for the date column	Identifies correct filter	Cannot locate filter without assistance
2	Applies "last 7 days" filter condition	Filter applied; row count reduces	Incorrect filter applied; incorrect row count
3	Identifies the number of submissions in the filtered view	States correct count	States incorrect count
4	Correctly identifies missing unit(s) by comparing submission list to provided unit roster	All missing units named	Misses a missing unit or incorrectly identifies a present unit as missing

T&EO TM10-03: EXECUTE AN AUTHORIZED ACTION

Task: Execute a status update Action on a specified record in a Workshop application.

Conditions: Given a Workshop application with a SITREP status table and an "Update SITREP Status" Action; the evaluator specifies the target record (by name or ID) and the new status value to set.

Standards: The trainee will locate the target record, execute the Action with the correct parameters, and verify the status updated — without assistance — within 3 minutes.

#	Performance Measure	GO	NO-GO
1	Locates the target record in the table (by filtering or scrolling)	Record located	Cannot locate record; requires assistance
2	Activates the Action button (selects record first if required)	Action button activated	Action button remains grayed; trainee does not diagnose
3	Completes the Action parameter form with the specified new status	Correct status value entered	Incorrect status entered
4	Confirms the Action execution when prompted	Executes Action	Dismisses confirmation without executing
5	Verifies the record status updated in the table	Confirms updated status visible	Does not verify; states "it worked" without confirmation
[CRITICAL] 6	Does not execute the Action on the wrong record	Target record status updated	Wrong record status updated

T&EO TM10-04: EXPORT FILTERED TABLE TO CSV

Task: Export a filtered Workshop table to CSV with correct classification labeling.

Conditions: Given a filtered Workshop table (filter previously applied); the evaluator specifies the export destination (a designated training output folder).

Standards: The trainee will export the filtered table to CSV, confirm row count matches the filtered view, and label the file with the correct classification marking.

#	Performance Measure	GO	NO-GO
1	Locates the export function for the table widget	Export function found	Cannot locate; requires assistance
2	Selects CSV as export format	CSV selected	Incorrect format selected
3	Exports to the designated training output folder (not personal downloads, cloud storage)	Exported to authorized location	Exported to unauthorized location
4	Confirms exported row count matches filtered table row count	Row count verified	Row count not verified; or count does not match
[CRITICAL] 5	Labels the exported file with the classification marking (UNCLASSIFIED in training)	File labeled	File not labeled; or incorrect label

T&EO TM10-05: BUILD A BASIC CONTOUR CHART

Task: Build a bar chart in Contour from a provided dataset.

Conditions: Given access to Contour and a specified dataset (equipment readiness by unit) in the Training Environment; the evaluator specifies the chart type (bar chart), X axis field (unit), and Y axis field (equipment_count); a filter condition is also specified (c_rating IN ['C3','C4']).

Standards: The trainee will open the dataset in Contour, build the specified bar chart, apply the specified filter, and describe what the chart shows — without assistance — within 10 minutes.

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#	Performance Measure	GO	NO-GO
1	Opens the specified dataset in Contour	Dataset open in Contour	Cannot open; navigates to wrong dataset
2	Selects correct X axis field (unit)	Correct field on X axis	Incorrect field; chart does not show unit distribution
3	Selects correct Y axis field (equipment_count or equivalent)	Correct field on Y axis	Incorrect field
4	Applies specified filter (c_rating IN ['C3','C4'])	Filter applied; data narrows	Filter not applied or incorrect
5	Saves the Contour analysis with a descriptive name	Analysis saved with descriptive name	Not saved; or named with default/non-descriptive label

T&EO TM10-06: IDENTIFY CLASSIFICATION MARKING AND STATE AUTHORIZED EXPORT PROCEDURE

Task: Locate the classification marking of a specified dataset and state the authorized export procedure.

Conditions: The evaluator identifies a dataset by name; the trainee must locate the classification marking in the dataset properties and state the authorized export destination and handling requirements for that classification level.

Standards: The trainee will locate the classification marking in the dataset Properties panel and correctly state the authorized distribution and export procedure — without reference to notes.

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#	Performance Measure	GO	NO-GO
1	Opens the dataset Properties panel	Properties panel open	Cannot locate Properties panel
2	Reads the classification marking from the Properties panel	Reads marking aloud	States a marking without reading it from the Properties panel
[CRITICAL] 3	States the correct authorized distribution for the marking	Correct distribution stated	States incorrect distribution (e.g., "FOUO data can go to personal email")
[CRITICAL] 4	States the authorized export destination (government systems for FOUO/CUI; no personal cloud storage, no personal email)	Correct destination stated	States an unauthorized destination
5	States the correct file labeling requirement for the export	Correct labeling stated	No labeling requirement stated

PART II — TM-20: BUILDER T&EOS

T&EO TM20-01: CREATE A FOUNDRY PROJECT TO STANDARD

Conditions: Given the C2DAO Naming and Governance Standards; a specified project name, classification, and folder structure.

Standards: Create a correctly named, marked, and structured project — independently.

#	Performance Measure	GO	NO-GO
1	Project name follows C2DAO naming convention (unit_description_env format)	Correct name	Spaces, special chars, or format violation
[CRITICAL] 2	Classification marking set	Marking set	No marking
3	All four required folders created (Datasets, Pipelines, Ontology, Applications)	All four present	Any folder missing

T&EO TM20-02: INGEST FILES AND VERIFY DATA QUALITY

Conditions: Two provided files (CSV and XLSX); trainee's Foundry project.

Standards: Ingest both files; verify row counts; identify at least one data quality observation per file.

#	Performance Measure	GO	NO-GO
1	Both files ingested to correct project folder	Both files in Datasets folder	File in wrong location or not ingested
2	Row count verified for each file (matches source)	Both row counts confirmed	Row count not checked
3	At least one data quality observation noted per file	Observation documented	No observation documented

T&EO TM20-03: BUILD A CLEAN-AND-TRANSFORM PIPELINE

Conditions: Two ingested datasets (equipment and unit lookup); trainee's project.

Standards: Build a pipeline that validates, cleans, type-casts, and joins the two datasets; pipeline runs to completion without error; output row count matches expected.

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#	Performance Measure	GO	NO-GO
1	Pipeline has Filter step removing null or invalid rows	Filter step present and configured	No filter; nulls present in output
2	Pipeline has Rename step with C2DAO-compliant column names	All output columns follow convention	Any column with non-compliant name
3	Pipeline has CAST steps correcting type errors identified in profiling	Types correct in output	Type mismatch in output
4	Pipeline has a Join step on `unit_id` producing a joined output	Join step present; correct key	No join; or wrong key; or 0-row output
5	Date column(s) have a computed DATEDIFF column where applicable	Date computation present and correct	Date computation absent or incorrect
[CRITICAL] 6	Pipeline runs to completion without error	No errors in pipeline log	Pipeline errors present (even if output appears)
7	Output row count matches expected (no unexplained fan-out)	Row count matches	Row count unexplainably inflated (fan-out)

T&EO TM20-04: CREATE AN OBJECT TYPE

Conditions: Pipeline output dataset with at least 5 columns of different types.

Standards: Create an Object Type with all specified properties correctly typed, a Primary Key, and a display name expression.

#	Performance Measure	GO	NO-GO
1	Object Type created with all required properties	All properties present	Any required property missing
[CRITICAL] 2	All property types correct (STRING, INTEGER, DATE, etc.)	All types correct	Any type incorrect
[CRITICAL] 3	Primary Key property designated	Primary Key set	No Primary Key
4	Display name expression configured	Display name expression set	No display name expression
5	Object Type naming follows C2DAO convention	Compliant name	Non-compliant name

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T&EO TM20-05: CREATE A LINK TYPE

Conditions: Two existing Object Types (Equipment and Unit).

Standards: Create a Link Type connecting the two Object Types with correct cardinality and directionality.

#	Performance Measure	GO	NO-GO
1	Link Type created between correct Object Types	Equipment links to Unit	Wrong Object Types
2	Cardinality set correctly (MANY_TO_ONE)	MANY_TO_ONE	Incorrect cardinality
3	Directionality correct (Equipment → Unit)	Correct direction	Reversed

T&EO TM20-06: CONFIGURE ONTOLOGY WRITE STEP

Conditions: Existing Object Type; pipeline with output dataset.

Standards: Configure an Ontology write step with correct property mapping; run pipeline; verify Object count in Quiver matches source row count.

#	Performance Measure	GO	NO-GO
1	Ontology write step added to pipeline	Write step present	No write step
2	All required properties mapped (pipeline column → Object property)	All properties mapped	Any required property unmapped
[CRITICAL] 3	Primary Key column mapped	PK mapped	PK not mapped (all objects overwrite)
[CRITICAL] 4	Object count in Quiver matches source row count	Count matches	Count does not match (silent data loss)

T&EO TM20-07: CONFIGURE AN ACTION

Conditions: Existing Object Type (Equipment).

Standards: Create an Action with a parameter, write rule, and Editor-only access restriction; test from Ontology Manager; confirm property updates.

#	Performance Measure	GO	NO-GO
1	Action created with named parameter	Parameter exists with name and type	No parameter
2	Write rule maps parameter to correct Object property	Write rule correct	Write rule missing or incorrect
[CRITICAL] 3	Access restricted to Editor role (Viewer cannot execute)	Editor-only restriction set	No restriction; Viewer can execute
4	Action tested from Ontology Manager; property confirms updated	Property updated after test	Property did not update

T&EO TM20-08: BUILD A WORKSHOP APPLICATION

Conditions: Object Type with at least 5 properties; existing Action.

Standards: Create a Workshop application with a table, filter widget, metric widget, and bar chart — each correctly bound to the Object Type.

#	Performance Measure	GO	NO-GO
1	Application named per C2DAO convention	Compliant name	Non-compliant name
2	Table widget bound to Object Type; displays required columns	Table shows live Ontology data	Table not bound; static or empty
3	Filter widget present and connected to table	Filter narrows table correctly	Filter present but not connected; or absent
4	Metric widget shows C1 count (or specified count)	Metric displays correct value	Metric absent or incorrect
5	Bar chart widget present with correct X and Y fields	Chart displays correct distribution	Chart absent; or wrong fields

T&EO TM20-09: CONNECT ACTION BUTTON; VERIFY EXECUTION

Conditions: Workshop application from T&EO TM20-08; Action from T&EO TM20-07.

Standards: Add an Action button; execute from a selected row; verify table refreshes with updated value.

#	Performance Measure	GO	NO-GO
1	Action button added to application	Button present	No button
2	Action executes when row is selected and button clicked	Action fires; confirmation appears	Action does not fire
[CRITICAL] 3	Table refreshes to show updated property value after execution	Table refreshes with correct updated value	Table does not refresh; or wrong value

T&EO TM20-10: CONFIGURE ACCESS CONTROL

Conditions: Workshop application; test Viewer-role account provided by evaluator.

Standards: Grant Viewer access to test account; confirm Viewer can see application and data but cannot execute Editor-only Action.

#	Performance Measure	GO	NO-GO
1	Test Viewer account granted access	Access granted	Access not granted
2	Viewer account can open and view the application	Application visible to Viewer	Application not visible
[CRITICAL] 3	Viewer account cannot execute the Editor-only Action	Action unavailable to Viewer	Viewer can execute Action (access misconfigured)

T&EO TM20-11: CREATE BRANCH AND SUBMIT PROMOTION REQUEST

Conditions: Workshop application to modify (change application header text as the branch change).

Standards: Create a branch; make the change on the branch; verify branch-only; submit promotion request with complete description; respond to evaluator (as data steward) feedback.

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#	Performance Measure	GO	NO-GO
[CRITICAL] 1	Branch created BEFORE making the change	Branch exists before change	Change made on main; branch created after
2	Change visible on branch; main is unchanged	Change is branch-only	Change appears on main
3	Promotion request submitted with complete description (what changed / why / downstream impact)	Description complete	Description empty or generic ("Updated application")
4	Evaluator feedback responded to within the evaluation period	Feedback addressed; revision submitted	Feedback ignored

PART III — TM-30: ADVANCED BUILDER T&EOS

T&EO TM30-01: DESIGN ONTOLOGY SCHEMA

Conditions: A provided mission scenario description; 30 minutes design time before tool access; 6-item design rubric.

Standards: Produce a documented Ontology schema (Object Types with properties and types, Link Types with cardinality, Actions with access restrictions) scoring ≥75% on the rubric with no zero-score item.

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#	Rubric Item	GO	NO-GO
1	Domain entities correctly identified (no missing required entities, no phantom entities)	All required entities present	Required entity missing; or phantom entity with no data basis
2	Primary Keys appropriate (unique per entity, non-null, stable identifier)	PK choices justified	PK choice clearly wrong (name field, unstable ID)
3	Property types documented and correct	All types specified and correct	Missing types; or significant type errors
4	Link Type cardinality correct and justified	Cardinality correct; rationale stated	Cardinality wrong; or no rationale
5	Actions target correct properties with appropriate access control	All Actions have access restriction specified	Any Action with no access restriction
6	Naming follows C2DAO convention	All names compliant	>2 naming violations

T&EO TM30-02: BUILD MULTI-SOURCE PIPELINE WITH APPEND MODE

Conditions: Two or three provided datasets; pipeline requirements specified in scenario.

Standards: Build a pipeline joining multiple sources; configure Append mode; run twice; verify two distinct snapshot records with different timestamps.

#	Performance Measure	GO	NO-GO
1	Join step on correct key; correct join type	Join correct	Wrong key; wrong join type; fan-out unresolved
[CRITICAL] 2	Fan-out detected and corrected if present	Fan-out absent; or detected and documented	Fan-out present in output without detection
3	Append mode configured before first run	Append mode set; confirmed before run	Overwrite mode used; or Append set after first run
4	Snapshot timestamp column present in output	Timestamp column present	No timestamp
[CRITICAL] 5	Two distinct snapshot records present after two runs	Two records with different timestamps	Only one record (Overwrite mode was used)

T&EO TM30-03: CONFIGURE APPEND MODE SNAPSHOT PIPELINE — *INCLUDED IN TM30-02 ABOVE*

T&EO TM30-04: BUILD MULTI-PAGE WORKSHOP APPLICATION

Conditions: Existing Object Types from scenario; scenario specifying two pages with variable passing.

Standards: Build a multi-page application where Page 1 object selection drives a filtered view on Page 2; conditional formatting on at least one table.

#	Performance Measure	GO	NO-GO
1	Page 1 (portfolio) shows all units with correct status	Portfolio page correct	Portfolio page empty or incorrect
2	Object selection on Page 1 navigates to Page 2	Navigation works	Navigation absent; or goes to wrong page
[CRITICAL] 3	Page 2 content filtered by Page 1 selection	Page 2 shows correct unit's records	Page 2 shows all records regardless of selection
4	Conditional formatting present (status-based row coloring)	Formatting applied	No conditional formatting

T&EO TM30-05: BUILD CONTOUR WORKBOOK WITH CALCULATED DEVIATION COLUMN

Standards: Build a Contour workbook showing readiness by battalion with a calculated deviation-from-standard column.

#	Performance Measure	GO	NO-GO
1	Correct dataset loaded	Correct dataset	Wrong dataset
2	Deviation column computed (actual - standard or equivalent)	Deviation column present and correct	Column absent or formula incorrect
3	Workbook saved with descriptive name	Saved with compliant name	Not saved

T&EO TM30-06: EXECUTE FULL C2DAO PROMOTION WORKFLOW

Standards: Branch → change → complete description → submit → respond to steward feedback.

#	Performance Measure	GO	NO-GO
[CRITICAL] 1	Promotion description complete (what changed/why/downstream impact)	Complete description	Empty or generic description
[CRITICAL] 2	Evaluator (data steward) feedback responded to	Feedback addressed; resubmission made	Feedback not addressed within evaluation
3	Change is on branch, not on main, at time of submission	Branch-only change	Change on main

PART IV — TM-40 SPECIALIST TRACK T&EOS

T&EOs for TM-40 specialist tracks are structured around the practical exercise tasks in each course syllabus. Evaluators should refer to the applicable course syllabus for the full Go standard and hard No-Go items. Critical items from each track are listed below.

TM-40A (ORSA) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
Time series forecast	90% confidence intervals present on forecast output	Yes — point estimate without bounds
Monte Carlo simulation	≥1,000 trials; seed set for reproducibility	Yes — fewer than 1,000 trials; no seed
Commander brief	Every estimate presented with confidence range	Yes — point estimate without bounds in brief
Monte Carlo reproducibility	Evaluator re-runs simulation; output matches	Yes — simulation is not reproducible

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TM-40B (AI ENGINEER) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
Human review gate	No AIP workflow write occurs without human checkpoint	Yes — write bypasses checkpoint
Authorization Checklist	Checklist complete and honest	Yes — checklist misrepresents workflow
Evaluator bypass attempt	Evaluator attempts to bypass review gate; trainee's workflow prevents it	Yes — bypass succeeds

TM-40C (MLE) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
Model calibration	Calibration check performed; documented	Yes — calibration skipped
Governance document	Model card includes: assumptions, training data description, limitations, intended use restrictions	Yes — required section missing
Drift detection	Evaluator-seeded drift event detected by monitoring pipeline	Yes — drift not detected

TM-40D (PROGRAM MANAGER) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
Dashboard data-as-of timestamp	Present on every commander-facing dashboard	Yes — absent
Append mode pipeline	Two distinct snapshot records after two runs	Yes — only one record
Contour portfolio sort	RED items sort to top (`overall_status` ascending)	No (affects task score, not hard No-Go)

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TM-40E (KNOWLEDGE MANAGER) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
AIP review gate	No AIP-generated lesson published without human review	Yes — auto-publish occurs
PCS transfer package	Names specific Foundry project, Object Types, pipelines, contacts	Yes — generic boilerplate

TM-40F (SOFTWARE ENGINEER) — CRITICAL PERFORMANCE MEASURES

Task	Critical Performance Measure	Hard NO-GO
Credential security	No hardcoded credentials in application code	Yes — any hardcoded credential
Validator test suite	All 8 test cases pass against trainee's actual code	Yes — any test case fails
Pagination	OSDK query iterates all pages (not just page 1)	No (affects task score)
Slate error states	Application displays useful error message on Action failure	No (affects task score)

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APPENDIX A — T&EO SCORING SHEET TEMPLATE

```
============================================================
MSS TRAINING T&EO SCORING SHEET
============================================================
Trainee Name:        ________________________________________
Course:              ________________________________________
Task/T&EO:           ________________________________________
Evaluation Date:     ________________________________________
Evaluator:           ________________________________________
============================================================

PERFORMANCE MEASURES

Step | Description                          | GO | NO-GO | N/A | Notes
-----|--------------------------------------|----|-------|-----|------
1    |                                      | [] | []    | []  |
2    |                                      | [] | []    | []  |
3    | [CRITICAL]                           | [] | []    | []  |
4    |                                      | [] | []    | []  |
5    |                                      | [] | []    | []  |

HARD NO-GO ITEMS VIOLATED (if any):
_____________________________________________________________

OVERALL TASK RESULT:  [ ] GO     [ ] NO-GO

EVALUATOR NOTES:
_____________________________________________________________
_____________________________________________________________

EVALUATOR SIGNATURE:  ________________________________________
DATE:                 ________________________________________
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